

Simulation study: mixture of experts model

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Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 3$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.35, 0.40, 0.25)$
- Degrees of freedom $\boldsymbol{\nu} = (8, 12, 3)$
- Scale matrices of the Wishart distribution $\Sigma_1, \Sigma_2, \Sigma_3$
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2) + \pi_3 \text{Wishart}(\nu_3, \Sigma_3)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(moewishart)

set.seed(5)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.35, 0.40, 0.25),
  nus = c(8, 12, 3),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 3

# run a moewishart model
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
fit <- moewishart(S_list,
  K = my_K_intial, mh_sigma = 0.2,
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)

Iter    1 | LL=-795.2 | acc_rate_nu=[0.000 0.000]
Iter  500 | LL=-745.7 | acc_rate_nu=[0.192 0.540]
Iter 1000 | LL=-731.7 | acc_rate_nu=[0.188 0.442]
Iter 1500 | LL=-735.2 | acc_rate_nu=[0.180 0.403]
Iter 2000 | LL=-730.4 | acc_rate_nu=[0.190 0.366]
```

```

Iter 2500 | LL=-740.4 | acc_rate_nu=[0.192 0.352]
Iter 3000 | LL=-738.8 | acc_rate_nu=[0.191 0.346]
Iter 3500 | LL=-733.1 | acc_rate_nu=[0.191 0.338]
Iter 4000 | LL=-734.7 | acc_rate_nu=[0.188 0.331]
Iter 4500 | LL=-735.4 | acc_rate_nu=[0.188 0.329]
Iter 5000 | LL=-732.5 | acc_rate_nu=[0.187 0.320]
Iter 5500 | LL=-733.0 | acc_rate_nu=[0.189 0.319]
Iter 6000 | LL=-733.8 | acc_rate_nu=[0.189 0.315]
Iter 6500 | LL=-734.0 | acc_rate_nu=[0.188 0.311]
Iter 7000 | LL=-730.6 | acc_rate_nu=[0.187 0.308]
Iter 7500 | LL=-733.0 | acc_rate_nu=[0.186 0.309]
Iter 8000 | LL=-731.9 | acc_rate_nu=[0.186 0.308]
Iter 8500 | LL=-734.6 | acc_rate_nu=[0.186 0.306]
Iter 9000 | LL=-733.2 | acc_rate_nu=[0.186 0.304]
Iter 9500 | LL=-741.4 | acc_rate_nu=[0.186 0.303]
Iter 10000 | LL=-735.5 | acc_rate_nu=[0.185 0.303]

```

```

end_time <- Sys.time()
elapsed_time <- end_time - start_time
elapsed_time

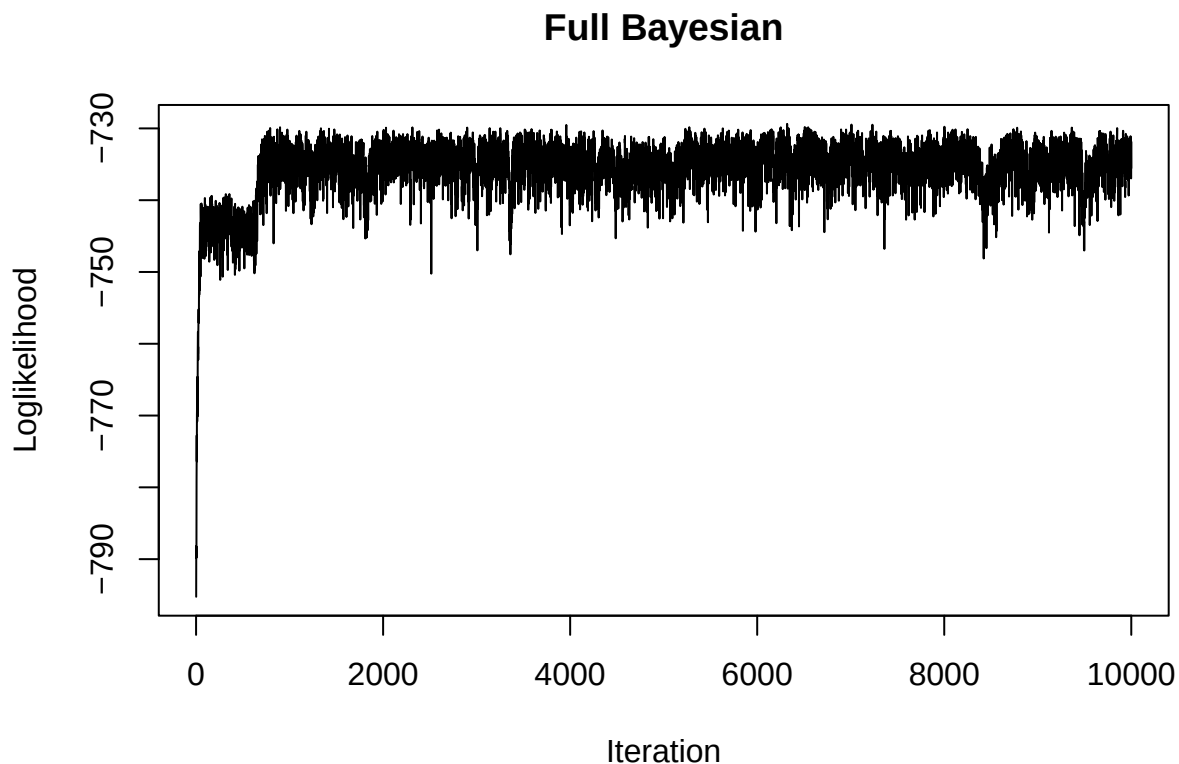
```

Time difference of 11.47069 secs

```

# likelihood
plot(fit$loglik, type = "l", lty = 1,
     main = "Full Bayesian",
     xlab = "Iteration", ylab = "Loglikelihood")

```



```

# posterior summaries
colMeans(fit$pi) # estimated pi

```

```
[1] 0.2421984 0.3876252 0.3701764
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit$nu, 2, mean) # estimated nu
```

```
[1] 12.485247 12.717498 3.409774
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```
# Sum all arrays in the list and divide by the number of saved iterations
```

```
sigma_posterior_mean <- Reduce("+", fit$Sigma) / length(fit$Sigma)
```

```
# View the mean covariance matrix for Cluster 1
```

```
# print(sigma_posterior_mean[, , 1])
```

```
# print(sigma_posterior_mean[, , 2])
```

```
# print(sigma_posterior_mean[, , 3])
```

```
# as.data.frame(Sigma_list)
```

```
(a1 <- apply(sigma_posterior_mean, MARGIN = 3, function(aa) min(sapply(Sigma_list, function(x) sum((aa - x)^2)))))
```

```
[1] 0.10220279 0.01779104 0.63433086
```

```
# run via Rcpp
```

```
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible  
set.seed(123)
```

```
fitMMcpp <- moewishart(S_list, K = 3, cpp = TRUE,
```

```
                    mh_sigma = 0.2,
```

```
                    niter = 3000, burnin = 1000, verbose = FALSE)
```

```
end_time <- Sys.time()
```

```
elapsed_time <- end_time - start_time
```

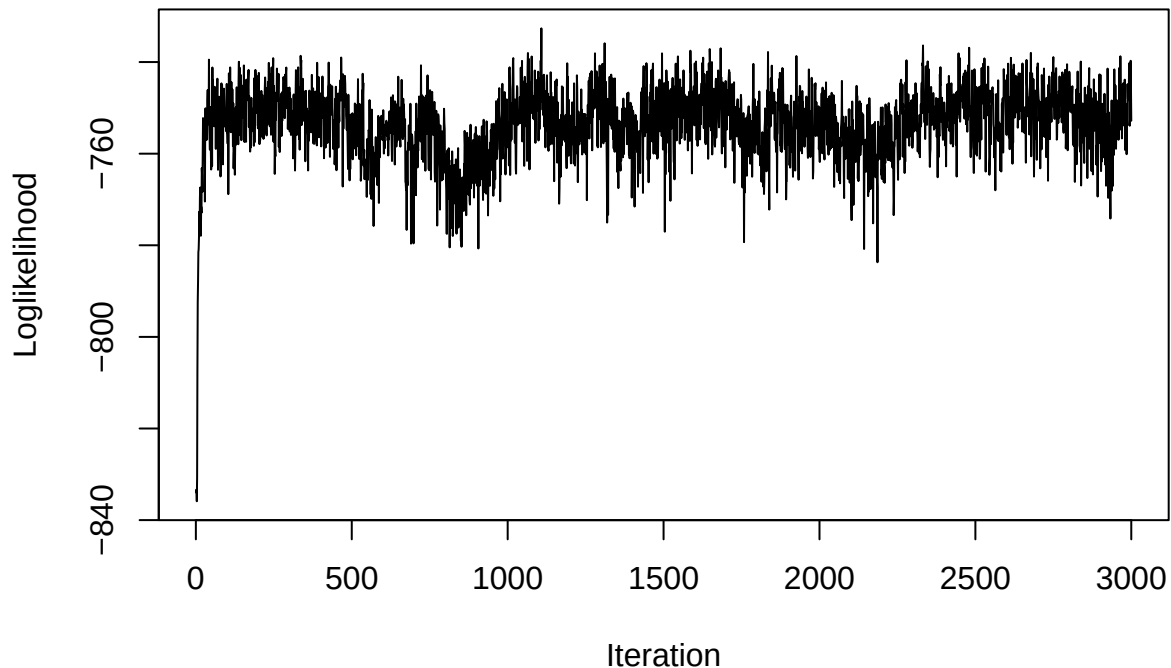
```
# likelihood
```

```
plot(fitMMcpp$loglik, type = "l", lty = 1,
```

```
     main = "Full Bayesian via Rcpp",
```

```
     xlab = "Iteration", ylab = "Loglikelihood")
```

Full Bayesian via Rcpp



```
# posterior summaries  
colMeans(fitMMcpp$pi) # estimated pi
```

```
[1] 0.3068362 0.3755280 0.3176358
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fitMMcpp$nu, 2, mean) # estimated nu
```

```
[1] 9.384792 3.407859 14.961991
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

Check identifiability issue

```
# re-run the model with wrong initial weights to check identifiability issue  
set.seed(1234)  
fit2 <- moewishart(S_list,  
  K = my_K_intial, init_pi = c(0.15, 0.40, 0.45), mh_sigma = 0.2,  
  niter = 3000, burnin = 1000, thin = 1, verbose = TRUE  
)
```

```
Iter    1 | LL=-795.1 | acc_rate_nu=[0.000 1.000]  
Iter   500 | LL=-732.8 | acc_rate_nu=[0.184 0.320]  
Iter  1000 | LL=-732.2 | acc_rate_nu=[0.176 0.306]  
Iter  1500 | LL=-733.0 | acc_rate_nu=[0.181 0.285]  
Iter  2000 | LL=-734.8 | acc_rate_nu=[0.188 0.286]  
Iter  2500 | LL=-732.7 | acc_rate_nu=[0.183 0.294]
```

```
Iter 3000 | LL=-732.8 | acc_rate_nu=[0.181 0.287]
```

```
colMeans(fit2$pi) # estimated pi
```

```
[1] 0.3735936 0.3640646 0.2623418
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
apply(fit2$nu, 2, mean) # estimated nu
```

```
[1] 13.875083 3.378355 11.269627
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```
# Sum all arrays in the list and divide by the number of saved iterations
```

```
sigma_posterior_mean2 <- Reduce("+", fit2$Sigma) / length(fit2$Sigma)
```

```
# View the mean covariance matrix for Cluster 1
```

```
print(sigma_posterior_mean2[, , 1])
```

```
      [,1]      [,2]
[1,] 1.8017513 0.5690862
[2,] 0.5690862 1.2697201
```

```
print(sigma_posterior_mean2[, , 2])
```

```
      [,1]      [,2]
[1,] 2.7187416 0.4107521
[2,] 0.4107521 2.1397348
```

```
print(sigma_posterior_mean2[, , 3])
```

```
      [,1]      [,2]
[1,] 0.3999361 0.1159827
[2,] 0.1159827 0.5019269
```

```
as.data.frame(Sigma_list)
```

```
      X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2 2.0 0.6 4.0 0.2
2 0.2 0.7 0.6 1.5 0.2 3.0
```

Bayesian MoE model with only intercepts

```
# run MoE model with only intercepts as covariates
```

```
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
```

```
fitMoE <- moewishartX(S_list, K = 3,
                      mh_sigma = 0.2, mh_beta = 0.4,
                      niter = 3000, burnin = 1000,
                      X = matrix(rep(1, n), ncol = 1))
```

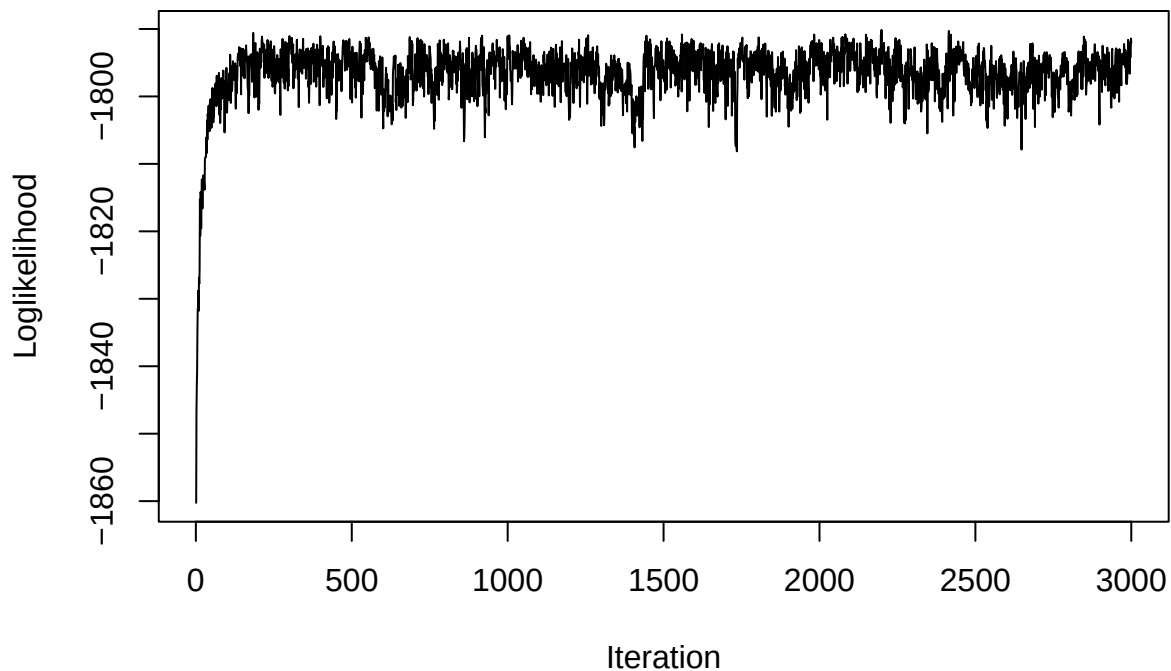
```
Iter    1 | LL=-1860.2 | acc_rate_nu=[1.000 1.000] | acc_rate_beta=0.000
Iter   500 | LL=-1794.5 | acc_rate_nu=[0.200 0.290] | acc_rate_beta=0.186
Iter  1000 | LL=-1791.0 | acc_rate_nu=[0.211 0.305] | acc_rate_beta=0.211
Iter  1500 | LL=-1797.4 | acc_rate_nu=[0.208 0.299] | acc_rate_beta=0.211
Iter  2000 | LL=-1794.5 | acc_rate_nu=[0.197 0.290] | acc_rate_beta=0.219
```

```
Iter 2500 | LL=-1796.3 | acc_rate_nu=[0.198 0.293] | acc_rate_beta=0.223
Iter 3000 | LL=-1792.1 | acc_rate_nu=[0.200 0.287] | acc_rate_beta=0.218
```

```
end_time <- Sys.time()
elapsed_time <- end_time - start_time

# likelihood
plot(fitMoE$loglik, type = "l", lty = 1,
     main = "MoE model with only intercept via R",
     xlab = "Iteration", ylab = "Loglikelihood")
```

MoE model with only intercept via R



```
#str(fitMoE)
# posterior summaries

colMeans(fitMoE$pi) # estimated pis

[1] 0.2445933 0.3854731 0.3699336

colMeans(dat$pi) # true pis

[1] 0.35 0.40 0.25

apply(fitMoE$nu, 2, mean) # estimated nu

[1] 11.633354 3.472146 13.323298

dat$nu # true nu

[1] 8 12 3

# Sum all arrays in the list and divide by the number of saved iterations
MoE_mean <- Reduce("+", fitMoE$Sigma) / length(fitMoE$Sigma)
# View the mean covariance matrix for Cluster 1
#print(MoE_mean[, , 1])
```

```
#print(MoE_mean[, , 2])
#print(MoE_mean[, , 3])
#as.data.frame(Sigma_list)
(a2 <- apply(MoE_mean, MARGIN = 3, function(aa) min(sapply(Sigma_list, function(x) sum((aa - x)^2))))))

[1] 0.07587089 0.70938923 0.04350780
```

Mixture model by EM algorithm

```
# run the mixture model with EM algorithm
set.seed(123)
fit3 <- moewishart(S_list,
  method = "em",
  K = my_K_intial, #init_pi = c(0.33, 0.33, 0.34),
  niter = 1000, verbose = TRUE
)
```

Running 3 initialization restarts...

```
-> Restart 1: Loglik = -1858.31
-> Restart 2: Loglik = -1858.64
-> Restart 3: Loglik = -1857.28
Iter 10 | Loglik: -1819.4271 | Nu: 16.45, 6.79, 5.26
Iter 20 | Loglik: -1813.5589 | Nu: 31.43, 5.8, 6.22
Iter 30 | Loglik: -1809.6575 | Nu: 46.42, 5.18, 7.2
Iter 40 | Loglik: -1807.1661 | Nu: 61.41, 4.83, 8.14
Iter 50 | Loglik: -1804.6516 | Nu: 76.4, 4.56, 9.22
Iter 60 | Loglik: -1802.4288 | Nu: 91.4, 4.4, 10.47
Iter 70 | Loglik: -1801.5065 | Nu: 106.39, 4.36, 11.33
Iter 80 | Loglik: -1801.1231 | Nu: 121.39, 4.34, 11.8
Iter 90 | Loglik: -1800.9006 | Nu: 136.39, 4.34, 12.04
Iter 100 | Loglik: -1800.7316 | Nu: 151.38, 4.33, 12.16
Iter 110 | Loglik: -1800.5863 | Nu: 166.38, 4.33, 12.22
Iter 120 | Loglik: -1800.4555 | Nu: 181.38, 4.33, 12.26
Iter 130 | Loglik: -1800.3357 | Nu: 196.38, 4.33, 12.27
Iter 140 | Loglik: -1800.2249 | Nu: 211.37, 4.33, 12.28
Iter 150 | Loglik: -1800.1218 | Nu: 226.37, 4.33, 12.28
Iter 160 | Loglik: -1800.0253 | Nu: 241.37, 4.33, 12.29
Iter 170 | Loglik: -1799.9346 | Nu: 256.37, 4.33, 12.29
Iter 180 | Loglik: -1799.8491 | Nu: 271.37, 4.33, 12.29
Iter 190 | Loglik: -1799.7682 | Nu: 286.37, 4.33, 12.29
Iter 200 | Loglik: -1799.6914 | Nu: 301.36, 4.33, 12.29
Iter 210 | Loglik: -1799.6184 | Nu: 316.36, 4.33, 12.29
Iter 220 | Loglik: -1799.5488 | Nu: 331.36, 4.33, 12.29
Iter 230 | Loglik: -1799.4822 | Nu: 346.36, 4.33, 12.29
Iter 240 | Loglik: -1799.4185 | Nu: 361.36, 4.33, 12.29
Iter 250 | Loglik: -1799.3574 | Nu: 376.36, 4.33, 12.29
Iter 260 | Loglik: -1799.2987 | Nu: 391.36, 4.33, 12.29
Iter 270 | Loglik: -1799.2422 | Nu: 406.36, 4.33, 12.29
Iter 280 | Loglik: -1799.1877 | Nu: 421.35, 4.33, 12.29
Iter 290 | Loglik: -1799.1352 | Nu: 436.35, 4.33, 12.29
Iter 300 | Loglik: -1799.0844 | Nu: 451.35, 4.33, 12.29
Iter 310 | Loglik: -1799.0353 | Nu: 466.35, 4.33, 12.29
Iter 320 | Loglik: -1798.9877 | Nu: 481.35, 4.33, 12.29
Iter 330 | Loglik: -1798.9416 | Nu: 496.35, 4.33, 12.29
```

Iter 340	Loglik: -1798.8969	Nu: 511.35, 4.33, 12.29
Iter 350	Loglik: -1798.8535	Nu: 526.35, 4.33, 12.29
Iter 360	Loglik: -1798.8113	Nu: 541.35, 4.33, 12.29
Iter 370	Loglik: -1798.7702	Nu: 556.35, 4.33, 12.29
Iter 380	Loglik: -1798.7303	Nu: 571.35, 4.33, 12.29
Iter 390	Loglik: -1798.6914	Nu: 586.35, 4.33, 12.29
Iter 400	Loglik: -1798.6534	Nu: 601.34, 4.33, 12.29
Iter 410	Loglik: -1798.6164	Nu: 616.34, 4.33, 12.29
Iter 420	Loglik: -1798.5803	Nu: 631.34, 4.33, 12.29
Iter 430	Loglik: -1798.5451	Nu: 646.34, 4.33, 12.29
Iter 440	Loglik: -1798.5106	Nu: 661.34, 4.33, 12.29
Iter 450	Loglik: -1798.4769	Nu: 676.34, 4.33, 12.29
Iter 460	Loglik: -1798.4440	Nu: 691.34, 4.33, 12.29
Iter 470	Loglik: -1798.4118	Nu: 706.34, 4.33, 12.29
Iter 480	Loglik: -1798.3802	Nu: 721.34, 4.33, 12.29
Iter 490	Loglik: -1798.3493	Nu: 736.34, 4.33, 12.29
Iter 500	Loglik: -1798.3191	Nu: 751.34, 4.33, 12.29
Iter 510	Loglik: -1798.2894	Nu: 766.34, 4.33, 12.29
Iter 520	Loglik: -1798.2603	Nu: 781.34, 4.33, 12.29
Iter 530	Loglik: -1798.2317	Nu: 796.34, 4.33, 12.29
Iter 540	Loglik: -1798.2037	Nu: 811.34, 4.33, 12.29
Iter 550	Loglik: -1798.1762	Nu: 826.34, 4.33, 12.29
Iter 560	Loglik: -1798.1492	Nu: 841.34, 4.33, 12.29
Iter 570	Loglik: -1798.1227	Nu: 856.34, 4.33, 12.29
Iter 580	Loglik: -1798.0966	Nu: 871.33, 4.33, 12.29
Iter 590	Loglik: -1798.0710	Nu: 886.33, 4.33, 12.29
Iter 600	Loglik: -1798.0458	Nu: 901.33, 4.33, 12.29
Iter 610	Loglik: -1798.0210	Nu: 916.33, 4.33, 12.29
Iter 620	Loglik: -1797.9966	Nu: 931.33, 4.33, 12.29
Iter 630	Loglik: -1797.9727	Nu: 946.33, 4.33, 12.29
Iter 640	Loglik: -1797.9491	Nu: 961.33, 4.33, 12.29
Iter 650	Loglik: -1797.9258	Nu: 976.33, 4.33, 12.29
Iter 660	Loglik: -1797.9029	Nu: 991.33, 4.33, 12.29
Iter 670	Loglik: -1797.8804	Nu: 1006.33, 4.33, 12.29
Iter 680	Loglik: -1797.8582	Nu: 1021.33, 4.33, 12.29
Iter 690	Loglik: -1797.8363	Nu: 1036.33, 4.33, 12.29
Iter 700	Loglik: -1797.8147	Nu: 1051.33, 4.33, 12.29
Iter 710	Loglik: -1797.7935	Nu: 1066.33, 4.33, 12.29
Iter 720	Loglik: -1797.7725	Nu: 1081.33, 4.33, 12.29
Iter 730	Loglik: -1797.7518	Nu: 1096.33, 4.33, 12.29
Iter 740	Loglik: -1797.7314	Nu: 1111.33, 4.33, 12.29
Iter 750	Loglik: -1797.7113	Nu: 1126.33, 4.33, 12.29
Iter 760	Loglik: -1797.6914	Nu: 1141.33, 4.33, 12.29
Iter 770	Loglik: -1797.6718	Nu: 1156.33, 4.33, 12.29
Iter 780	Loglik: -1797.6525	Nu: 1171.33, 4.33, 12.29
Iter 790	Loglik: -1797.6334	Nu: 1186.33, 4.33, 12.29
Iter 800	Loglik: -1797.6145	Nu: 1201.33, 4.33, 12.29
Iter 810	Loglik: -1797.5959	Nu: 1216.33, 4.33, 12.29
Iter 820	Loglik: -1797.5775	Nu: 1231.33, 4.33, 12.29
Iter 830	Loglik: -1797.5594	Nu: 1246.32, 4.33, 12.29
Iter 840	Loglik: -1797.5414	Nu: 1261.32, 4.33, 12.29
Iter 850	Loglik: -1797.5237	Nu: 1276.32, 4.33, 12.29
Iter 860	Loglik: -1797.5061	Nu: 1291.32, 4.33, 12.29
Iter 870	Loglik: -1797.4888	Nu: 1306.32, 4.33, 12.29


```

Iter 880 | Loglik: -1797.4717 | Nu: 1321.32, 4.33, 12.29
Iter 890 | Loglik: -1797.4547 | Nu: 1336.32, 4.33, 12.29
Iter 900 | Loglik: -1797.4380 | Nu: 1351.32, 4.33, 12.29
Iter 910 | Loglik: -1797.4214 | Nu: 1366.32, 4.33, 12.29
Iter 920 | Loglik: -1797.4050 | Nu: 1381.32, 4.33, 12.29
Iter 930 | Loglik: -1797.3888 | Nu: 1396.32, 4.33, 12.29
Iter 940 | Loglik: -1797.3728 | Nu: 1411.32, 4.33, 12.29
Iter 950 | Loglik: -1797.3569 | Nu: 1426.32, 4.33, 12.29
Iter 960 | Loglik: -1797.3412 | Nu: 1441.32, 4.33, 12.29
Iter 970 | Loglik: -1797.3257 | Nu: 1456.32, 4.33, 12.29
Iter 980 | Loglik: -1797.3103 | Nu: 1471.32, 4.33, 12.29
Iter 990 | Loglik: -1797.2951 | Nu: 1486.32, 4.33, 12.29
Iter 1000 | Loglik: -1797.2800 | Nu: 1501.32, 4.33, 12.29

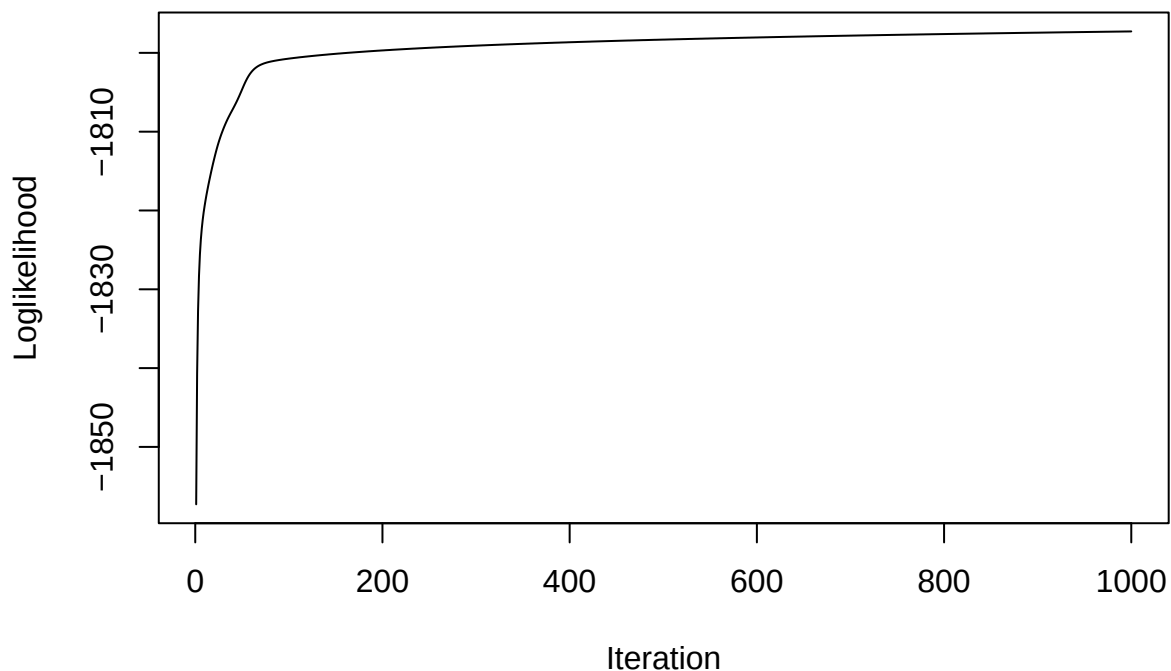
```

```

# likelihood
plot(fit3$loglik, type = "l", lty = 1,
     main = "EM algorithm via R",
     xlab = "Iteration", ylab = "Loglikelihood")

```

EM algorithm via R



```

# EM estimates
fit3$pi

```

```
[1] 0.004999943 0.601476939 0.393523118
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
fit3$nu
```

```
[1] 1502.819475    4.330382   12.287820
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```

```
# run the mixture model with Rcpp version of EM algorithm
```

```
set.seed(123)
```

```
fit3cpp <- moewishart(S_list,  
  method = "em",  
  K = my_K_intial, cpp = TRUE,  
  niter = 1000, verbose = TRUE  
)
```

```
Mixture-Wishart init: n=200 p=2 K=3
```

```
EM iter 0, loglik=-2402.35
```

```
Updated nu[0] from 4 to 8.58952
```

```
Updated nu[1] from 4 to 8.58506
```

```
Updated nu[2] from 4 to 8.59073
```

```
EM iter 1, loglik=-2456.43
```

```
Updated nu[0] from 8.58952 to 8.59572
```

```
Updated nu[1] from 8.58506 to 8.56553
```

```
Updated nu[2] from 8.59073 to 8.60442
```

```
EM iter 2, loglik=-2061.55
```

```
Updated nu[0] from 8.59572 to 8.64695
```

```
Updated nu[1] from 8.56553 to 8.40841
```

```
Updated nu[2] from 8.60442 to 8.71809
```

```
EM iter 3, loglik=-2058.88
```

```
Updated nu[0] from 8.64695 to 9.11076
```

```
Updated nu[1] from 8.40841 to 7.31851
```

```
Updated nu[2] from 8.71809 to 9.74323
```

```
EM iter 4, loglik=-1954.4
```

```
Updated nu[0] from 9.11076 to 12.7766
```

```
Updated nu[1] from 7.31851 to 5.23653
```

```
Updated nu[2] from 9.74323 to 16.6466
```

```
EM iter 5, loglik=-1899.84
```

```
Updated nu[0] from 12.7766 to 15.9581
```

```
Updated nu[1] from 5.23653 to 5.19171
```

```
Updated nu[2] from 16.6466 to 22.71
```

```
EM iter 6, loglik=-1841.16
```

```
Updated nu[0] from 15.9581 to 18.1618
```

```
Updated nu[1] from 5.19171 to 5.5459
```

```
Updated nu[2] from 22.71 to 26.0107
```

```
EM iter 7, loglik=-1831.14
```

```
Updated nu[0] from 18.1618 to 19.1435
```

```
Updated nu[1] from 5.5459 to 5.80559
```

```
Updated nu[2] from 26.0107 to 27.8551
```

```
EM iter 8, loglik=-1832.69
```

```
Updated nu[0] from 19.1435 to 19.3583
```

```
Updated nu[1] from 5.80559 to 5.87436
```

```
Updated nu[2] from 27.8551 to 29.5079
```

```
EM iter 9, loglik=-1832.43
```

```
Updated nu[0] from 19.3583 to 19.2466
```

```
Updated nu[1] from 5.87436 to 5.83442
```

```
Updated nu[2] from 29.5079 to 30.9826
```

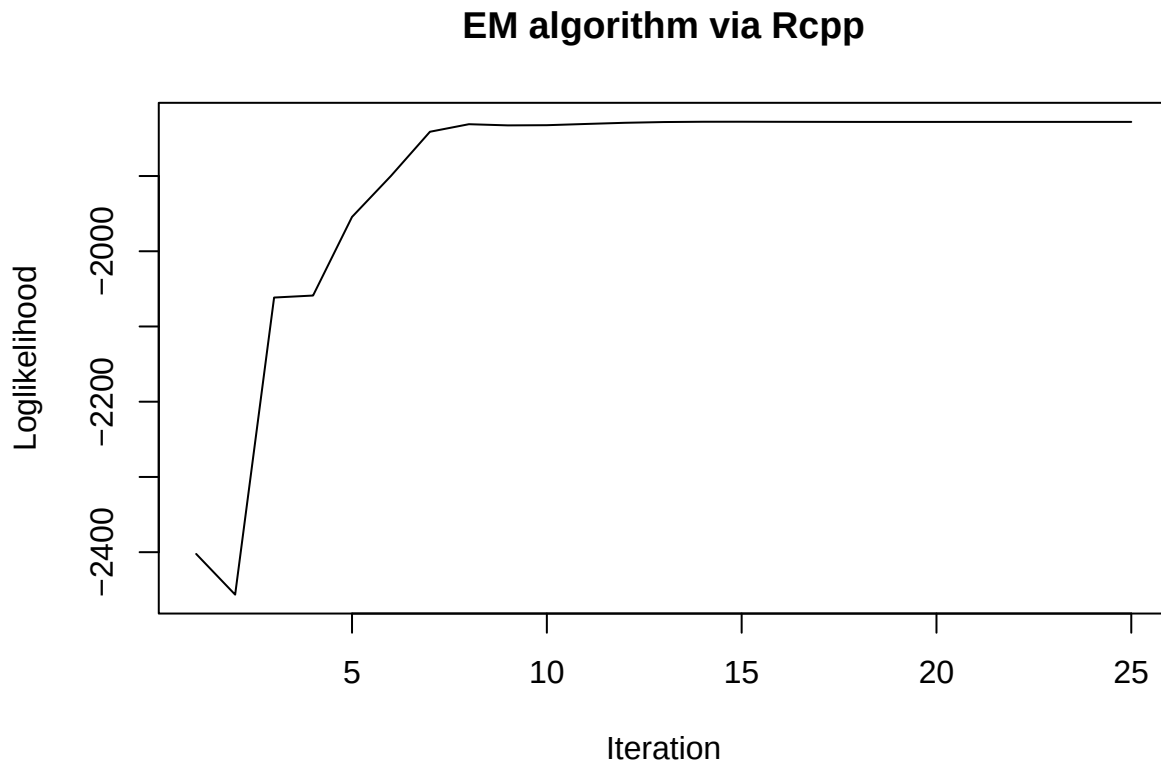
```
EM iter 10, loglik=-1830.81
```

```
Updated nu[0] from 19.2466 to 19.0792
```

Updated nu[1] from 5.83442 to 5.77321
 Updated nu[2] from 30.9826 to 31.8898
 EM iter 11, loglik=-1829.25
 Updated nu[0] from 19.0792 to 18.9655
 Updated nu[1] from 5.77321 to 5.73179
 Updated nu[2] from 31.8898 to 32.1948
 EM iter 12, loglik=-1828.28
 Updated nu[0] from 18.9655 to 18.9184
 Updated nu[1] from 5.73179 to 5.71502
 Updated nu[2] from 32.1948 to 32.1409
 EM iter 13, loglik=-1827.89
 Updated nu[0] from 18.9184 to 18.9131
 Updated nu[1] from 5.71502 to 5.71357
 Updated nu[2] from 32.1409 to 31.9714
 EM iter 14, loglik=-1827.85
 Updated nu[0] from 18.9131 to 18.9233
 Updated nu[1] from 5.71357 to 5.71767
 Updated nu[2] from 31.9714 to 31.8199
 EM iter 15, loglik=-1827.94
 Updated nu[0] from 18.9233 to 18.9333
 Updated nu[1] from 5.71767 to 5.72169
 Updated nu[2] from 31.8199 to 31.7258
 EM iter 16, loglik=-1828.03
 Updated nu[0] from 18.9333 to 18.9384
 Updated nu[1] from 5.72169 to 5.72384
 Updated nu[2] from 31.7258 to 31.68
 EM iter 17, loglik=-1828.08
 Updated nu[0] from 18.9384 to 18.9392
 Updated nu[1] from 5.72384 to 5.72436
 Updated nu[2] from 31.68 to 31.6603
 EM iter 18, loglik=-1828.09
 Updated nu[0] from 18.9392 to 18.9378
 Updated nu[1] from 5.72436 to 5.72405
 Updated nu[2] from 31.6603 to 31.6494
 EM iter 19, loglik=-1828.08
 Updated nu[0] from 18.9378 to 18.936
 Updated nu[1] from 5.72405 to 5.72357
 Updated nu[2] from 31.6494 to 31.6385
 EM iter 20, loglik=-1828.07
 Updated nu[0] from 18.936 to 18.9345
 Updated nu[1] from 5.72357 to 5.72319
 Updated nu[2] from 31.6385 to 31.6254
 EM iter 21, loglik=-1828.06
 Updated nu[0] from 18.9345 to 18.9336
 Updated nu[1] from 5.72319 to 5.72298
 Updated nu[2] from 31.6254 to 31.6108
 EM iter 22, loglik=-1828.06
 Updated nu[0] from 18.9336 to 18.9329
 Updated nu[1] from 5.72298 to 5.72288
 Updated nu[2] from 31.6108 to 31.5963
 EM iter 23, loglik=-1828.05
 Updated nu[0] from 18.9329 to 18.9325
 Updated nu[1] from 5.72288 to 5.72283
 Updated nu[2] from 31.5963 to 31.5827

```
EM iter 24, loglik=-1828.05
Converged at iter 24, loglik = -1828.05, rel = 8.60687e-07
```

```
# likelihood
plot(fit3cpp$loglik, type = "l", lty = 1,
     main = "EM algorithm via Rcpp",
     xlab = "Iteration", ylab = "Loglikelihood")
```



```
# EM estimates
fit3cpp$pi
```

```
      [,1]
[1,] 0.352911691
[2,] 0.638486008
[3,] 0.008602301
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
fit3cpp$nu
```

```
      [,1]
[1,] 18.932451
[2,]  5.722833
[3,] 31.582676
```

```
dat$nu # true nu
```

```
[1]  8 12  3
```

MoE model with only intercepts by EM algorithm

```
# run the MoE model with EM algorithm
set.seed(123)
fit3MoE <- moewishartX(S_list,
  method = "em", X = matrix(rep(1, n), ncol = 1),
  K = my_K_intial, #init_pi = c(0.33, 0.33, 0.34),
  niter = 3000, verbose = TRUE
)
```

```
Iter   1  loglik = -1976.244601
Iter   2  loglik = -1883.753623
Iter   3  loglik = -1869.123347
Iter   4  loglik = -1839.451255
Iter   5  loglik = -1813.479546
Iter   6  loglik = -1807.817409
Iter   7  loglik = -1807.126605
Iter   8  loglik = -1805.747114
Iter   9  loglik = -1802.393795
Iter  10  loglik = -1797.826526
Iter  11  loglik = -1794.201057
Iter  12  loglik = -1792.137979
Iter  13  loglik = -1791.093824
Iter  14  loglik = -1790.514189
Iter  15  loglik = -1790.136551
Iter  16  loglik = -1789.862114
Iter  17  loglik = -1789.650995
Iter  18  loglik = -1789.484100
Iter  19  loglik = -1789.350795
Iter  20  loglik = -1789.244255
Iter  21  loglik = -1789.159475
Iter  22  loglik = -1789.092440
Iter  23  loglik = -1789.039790
Iter  24  loglik = -1788.998696
Iter  25  loglik = -1788.966800
Iter  26  loglik = -1788.942158
Iter  27  loglik = -1788.923195
Iter  28  loglik = -1788.908648
Iter  29  loglik = -1788.897518
Iter  30  loglik = -1788.889021
Iter  31  loglik = -1788.882545
Iter  32  loglik = -1788.877616
Iter  33  loglik = -1788.873869
Iter  34  loglik = -1788.871022
Iter  35  loglik = -1788.868861
Iter  36  loglik = -1788.867222
Iter  37  loglik = -1788.865979
Iter  38  loglik = -1788.865037
Iter  39  loglik = -1788.864323
Iter  40  loglik = -1788.863782
Iter  41  loglik = -1788.863373
Iter  42  loglik = -1788.863063
Iter  43  loglik = -1788.862828
Iter  44  loglik = -1788.862650
```

```

Iter 45 loglik = -1788.862515
Iter 46 loglik = -1788.862413
Iter 47 loglik = -1788.862341
Iter 48 loglik = -1788.862265
Iter 49 loglik = -1788.862223
Iter 50 loglik = -1788.862193
Iter 51 loglik = -1788.862168
Iter 52 loglik = -1788.862153
Iter 53 loglik = -1788.862140
Iter 54 loglik = -1788.862131
Iter 55 loglik = -1788.862125
Iter 56 loglik = -1788.862118
Iter 57 loglik = -1788.862114
Iter 58 loglik = -1788.862111
Iter 59 loglik = -1788.862107
Iter 60 loglik = -1788.862105
Iter 61 loglik = -1788.862104
Iter 62 loglik = -1788.862102
Iter 63 loglik = -1788.862101
Iter 64 loglik = -1788.862100
Converged by loglik tolerance.

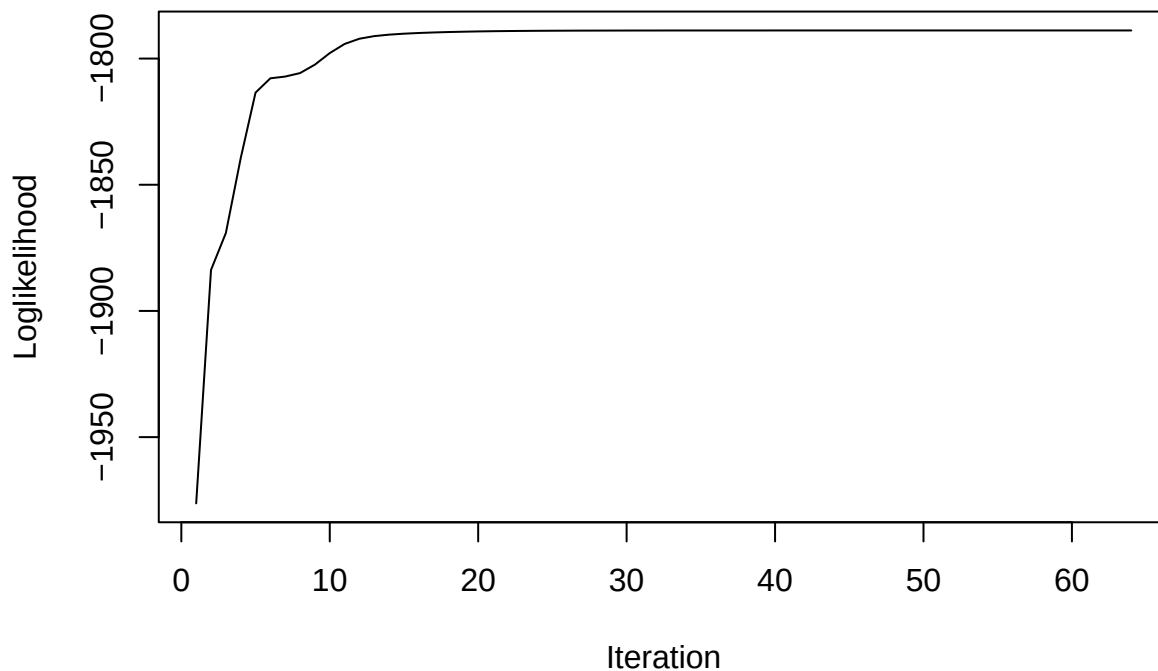
```

```

# likelihood
plot(fit3MoE$loglik, type = "l", lty = 1,
     main = "MoE model with only intercept by EM algorithm via R",
     xlab = "Iteration", ylab = "Loglikelihood")

```

MoE model with only intercept by EM algorithm via R



```

# EM estimates
colMeans(fit3MoE$gamma)

```

```
[1] 0.3546358 0.3868617 0.2585025
```

```
colMeans(dat$pi) # true pi
```

```
[1] 0.35 0.40 0.25
```

```
fit3MoE$nu
```

```
[1] 3.258293 12.598654 11.647128
```

```
dat$nu # true nu
```

```
[1] 8 12 3
```